# Dmitrii **Altukhov**

Senior Python AI Engineer · PhD (Computer Science)

*Amsterdam, The Netherlands*

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## Profile

ML Engineer with a PhD in computer science built around numerical optimization and a software engineering track record of building high-performance algorithms and scalable production AI systems. Background bridges algorithm-level performance engineering — CUDA/cuBLAS, C/C++, vectorized numerical methods, 10x algorithmic speedups — with distributed, event-driven production infrastructure. Strong foundations in linear algebra, optimization, and signal processing.

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| **Performance & GPU** | CUDA, cuBLAS, C/C++, numerical optimization, vectorization |
| **Languages** | Python, C/C++, Go, Rust (learning) |
| **ML** | PyTorch, deep learning, evaluation, LLM integration |
| **Infrastructure** | Kubernetes, Docker, Kafka, Redis, AWS, CI/CD, SQL |
| **Quantitative** | algorithms, linear algebra, statistics, signal processing |

## Experience

### Creative Fabrica — *Amsterdam, the Netherlands*

ML Engineer → Senior ML Engineer — *Dec. 2023 – present*

- Developed an advanced color-quantization algorithm that became foundational for an industry-leading image vectorization pipeline.
- Own the generative AI **Python** codebase, infrastructure, and internal testing and QA tool.
- Redesigned the architecture for image generation backend on **AWS**, significantly improving scalability and system performance.
- Designed and maintained distributed event-driven architecture using **Kafka**, **Redis** queues, and **Kubernetes**.
- Developed and deployed to production an inference pipeline for content matching.

### Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained a deep learning model to predict speech from brain data in **PyTorch**.
- Developed real-time **Python** pipeline for EEG timeseries prediction using Kalman filtering.

### Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations.
- Accelerated various parts of my PhD codebase to enable Monte-Carlo statistics, using **CUDA** (cuBLAS) and **C**.
- Led 4-people group on a project for real-time ML feature extraction and GPU-accelerated 3D visualization of brain activity in **Python**.

### University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Co-developed an open-source **Python** package for GPU-accelerated brain data visualization Visbrain, Combrisson et al. [2019].

### Scientific Research Institute of System Analysis (SRISA RAS) — *Moscow, Russia*

Research assistant → Junior research fellow — *Jun. 2011 – Jan. 2015*

- Developed a droplet evaporation model in **C++**; tested **CUDA**-accelerated cluster software for simulations of flow in jet engines.

## Education

### Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

GPA: 4.0; proposed an algorithmic framework for brain connectivity estimation; research published in the leading domain journals.

### Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

Thesis: “Enhancement and validation of LOGOS software for simulations of the reactive fluid flows.”